

WILL MCGOWAN

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EDUCATION

University of Strathclyde, Glasgow

Sep 2023 - Jun 2027 (expected)

BA (Hons) Economics

- Average grade to date: 2:1

- **Relevant Modules:** Applied Econometrics; Treasury Management and Derivatives; Advanced Microeconomics

Dissertation | MATLAB

In progress, due 2027

"Does the News Pay? Forecasting European Gas Volatility with MIDAS, Storage and Text Data"

- Testing whether news attention improves forecasts of TTF gas price volatility with mixed-frequency (MIDAS) models built on daily storage, market and news data, including an original Google Trends attention index
- Benchmarking out-of-sample against HAR and AR models with Diebold-Mariano tests before, during and after the crisis, and measuring economic value through the Sharpe ratio of a volatility-targeted TTF futures strategy

Belfast Royal Academy, Belfast

Jun 2023

- A-levels: Mathematics (A), Economics (A), Computer Science (B)

PROJECTS

Plains Heat and the KC-Chicago Wheat Spread

R

- Tested whether Plains heatwaves move Kansas City wheat futures against Chicago, using 27 summers of NOAA weather data and the off-season as a placebo: heat explained 0.3% of weekly moves
- Found a one-day timestamp error in the price data that would have faked a lead-lag signal; with it fixed, a trading rule set in advance showed no edge out of sample (2010-23)

Delta-Hedging Simulator

MATLAB

- Simulated delta-hedging a short call on 50,000 paths: hedging four times as often halved the hedging error, and 0.1-0.5% trading costs moved the best frequency from daily to weekly
- Tested reheding only when the hedge drifts a set band from delta: on fresh paths it cut RMS P&L by 6% against the best timetable at 0.1% costs, shrinking to 2% at 0.5%

Monte Carlo Option Pricing Engine

MATLAB

- Priced an arithmetic Asian call by Monte Carlo, checked against two closed-form prices; a geometric-Asian control variate cut the variance 1,276-fold (97% off the standard error)
- Mapped where the method stops working over 8 volatilities and 6 strikes: at the money the gain falls from 4,569x to 71x as volatility rises from 10% to 80%

Empirical CAPM Test

MATLAB

- Tested the CAPM on 694 anomaly portfolios in 54 countries (JKP data): alpha significant in 31% at 5%, over six times the rate expected by chance; GRS rejected in 48 of 53 countries
- Corrected for running 694 tests at once: 123 of the 215 significant alphas survive Benjamini-Hochberg and 84 clear $|t| > 3$; momentum holds up best, size vanishes

WORK EXPERIENCE

Le Set, Glasgow | Head Mixologist

Sep 2026 - Present

- Wrote a new 200-cover venue's 11-drink menu, including 4 original cocktails; trained around 15 bar staff

KONG, Glasgow | Mixologist and Front of House

Oct 2024 - Jun 2026

- Analysed sales patterns to refine upselling prompts, raising average customer spend by an estimated 25%; reconciled cash and card takings at the end of each shift

LEADERSHIP AND ACTIVITIES

Strathclyde Agricapital Society | Co-Founder and Treasurer

Sep 2025 - Present

- Co-founded Strathclyde's first agri-finance society, grew it to 126 members and published weekly research notes

Strathclyde Poker Society | Vice-President, then President

Sep 2025 - Apr 2026

- Ran weekly tournaments for around 60 players on £600 a semester, reconciling every buy-in and prize pool

ADDITIONAL INFORMATION

Technical: R, MATLAB, Excel, Git

Interests: poker, over £40,000 in lifetime cash-game profit, using the Kelly criterion to pick buy-in levels and manage bankroll (minimum 50 buy-ins); chess, 14th at the Irish U19 Chess Championships (2023) and won 21 of 22 games in a simultaneous exhibition against players averaging 1860 FIDE; powerlifting, Northern Ireland U19 champion (2023)